MedAuth-VLM: Multimodal Medical Vision-Language Architecture for Prior Authorization Evidence Extraction, Cross-Modal Grounding, and Policy Verification

Author: Nilesh Marathe

Project: MedAuth-VLM

Affiliation: Medical AI Systems Laboratory

Monograph Edition: Technical Research Book (Complete Monograph Release)

Date: October 2026

Repository State: Verified & Reproducible (PostgreSQL 18.6 Live Ingestion)

Front Matter

Abstract

Prior authorization (PA) represents one of the most resource-intensive administrative bottlenecks in modern healthcare systems, consuming an estimated 31 billion annually in administrative overhead and introducing clinical care delays for over 90% of practicing physicians. In diagnostic thoracic imaging, utilization management requires medical directors to manually cross-reference complex, unstructured clinical orders, radiologist dictations, and high-resolution radiograph pixels against state-specific Centers for Medicare & Medicaid Services (CMS) coverage guidelines. Conventional automated approaches suffer from unimodal siloing, generative hallucinations, and the absence of deterministic clinical grounding safeguards.

This monograph presents **MedAuth-VLM**, an end-to-end multimodal Vision-Language Model (VLM) architecture and deterministic clinical policy verification framework designed to automate precomputed evidence extraction, enforce cross-modal grounding consistency, and streamline physician-adjudicated prior authorization workflows. Centered upon Google's instruction-tuned **MedGemma-4B-IT** foundation model, MedAuth-VLM processes dual-projection chest radiographs (Frontal PA/AP and Lateral) alongside dictated radiology reports to extract structured, standardized clinical evidence subobjects across anatomical body parts, primary findings, specific abnormalities, text-image consistency indicators, and clinical support rationales.

We document the architectural design, database modeling, schema normalization, and empirical distribution of an extensive multimodal corpus comprising **2,107 clinical cases** (1,054 normal, 1,053 abnormal), **4,093 high-resolution radiograph projections** (2,099 frontal, 1,994 lateral), **2,107 structured JSONB evidence subobjects**, and **100 CMS diagnostic imaging insurance policies** persisted within an ACID-compliant PostgreSQL 18 repository. (An additional 43 uncomputed Batch 5 cases were preserved in recovery manifests following Colab GPU timeouts and strictly excluded from database ingestion and evaluation). We demonstrate how MedAuth-VLM enforces administrative clinical grounding safeguards through a deterministic policy comparison engine that restricts compliance scoring to ≤ 40/100 whenever source clinical documentation is absent (n = 22 cases, 1.04% of corpus), mandating human escalation (this policy cap represents a software safety guardrail, not proof of clinical correctness or hallucination elimination).

Crucially, in strict adherence to scientific integrity protocols, real-data diagnostic performance across the 2,107 cases is evaluated strictly as an exploratory engineering evaluation against documented dataset-derived heuristic reference labels (Indiana University Open-i MeSH indexing, Problems == 'normal' vs. abnormal terms). Against these reference labels, MedGemma-4B-IT abnormality token extraction demonstrates high abnormality specificity of 98.67% [95% CI: 97.78%, 99.21%] (1,040/1,054) and precision of 96.57% [95% CI: 94.32%, 97.95%] (394/408), alongside moderate sensitivity of 37.42% [95% CI: 34.54%, 40.38%] (394/1,053), ordinary accuracy of 68.06% [95% CI: 66.04%, 70.02%], balanced accuracy of 68.04%, and negative predictive value (NPV; corresponding to normal-class precision) of 61.21% [95% CI: 58.87%, 63.50%] (659 false negatives resulting from subtle findings documented in narrative prose rather than discrete token lists). We explicitly document dataset limitations, including an automated label heuristic, 39 unindexed MeSH cases mapped to abnormal, artificial 50/50 cohort balancing, and 400 Batch 6 cases with omitted raw markdown logging. System clinical diagnostic efficacy, real-world payer authorization accuracy, self-reported image-text consistency, spatial bounding-box grounding, and continuous ROC-AUC/ECE calibration are formally designated as NOT VALIDATED or NOT EVALUATED.

Keywords: Medical Vision-Language Models, MedGemma, Prior Authorization, Multimodal Cross-Modal Grounding, Chest Radiography, Utilization Management, Responsible AI, Clinical Decision Support.

Table of Contents

1. Chapter 1: Introduction

- 1.1 Healthcare Prior-Authorization Problem & Administrative Bottleneck
- 1.2 Motivation, Scope, and Research Questions
- 1.3 Intended Users & Clinical Use Cases
- 1.4 Project Contributions

2. Chapter 2: Literature Review and Background
• 2.1 Multimodal Vision-Language Models in Medicine
• 2.2 Medical Image Understanding in Thoracic Radiography
• 2.3 Clinical Text Processing & Radiology Information Extraction
• 2.4 Multimodal Evidence Matching & Cross-Modal Grounding
• 2.5 Healthcare Prior Authorization & Utilization Management
• 2.6 Responsible AI, Hallucination Mitigation & Human Oversight
• 2.7 Research Gap Analysis
3. Chapter 3: Problem Formulation and Requirements
• 3.1 Mathematical Problem Formulation
• 3.2 Assumptions & Operational Constraints
• 3.3 Functional Requirements
• 3.4 Non-Functional Requirements
• 3.5 Clinical Safety, Privacy & Governance Requirements
4. Chapter 4: System Architecture and Methodology
• 4.1 Four-Tier Decoupled Architecture
• 4.2 End-to-End Request & Data Flow
• 4.3 Subsystem Responsibilities
• 4.4 Architectural Design Decisions & Trade-Offs
• 4.5 Exception Handling, Failure Modes & Safeguards
5. Chapter 5: Dataset and Data Preparation
• 5.1 Clinical Cohorts & Insurance Policy Sources
• 5.2 Case & Projection Structure
• 5.3 Ground-Truth Derivation, Heuristics, and Limitations
• 5.4 Tripartite Evidence JSONB Schema
• 5.5 Batch Ingestion Composition (Batches 1–6)
• 5.6 Cryptographic SHA-256 Immutability Verification
• 5.7 Data Quality Checks, Missing Outputs & Provenance
6. Chapter 6: Models, Algorithms, and Implementation
• 6.1 Vision-Language Model Foundation: google/medgemma-4b-it
• 6.2 Offline GPU Precomputation Pipeline
• 6.3 Constrained Generation Prompting & JSON Schema Enforcement
• 6.4 Prediction Extraction & Decision Rule
• 6.5 Deterministic Policy Engine Logic
• 6.6 Source Code Explanations & Snippets
• 6.7 Runtime Environment & Dependencies
7. Chapter 7: Experimental Design and Evaluation Metrics
• 7.1 Empirical Research Questions
• 7.2 Evaluation Protocol: Operational Fidelity vs Diagnostic Efficacy
• 7.3 Mathematical Metric Formulations & Symbol Definitions
• 7.4 Metric Status Categorization Framework
• 7.5 Calculation Traceability Framework
• 7.6 Evaluation Readiness & Reference Standards Matrix
8. Chapter 8: Results and Detailed Analysis — PRIORITY CHAPTER
• 8.1 Evaluated Dataset & Corpus Summary Statistics
• 8.2 Batch-Wise Composition & Structural Breakdown
• 8.3 Evidence JSONB Completeness & Subobject Audit
• 8.4 Clinical Dictation Grounding Audit (The 22 Ungrounded Cases)
• 8.5 Deterministic Policy Engine Simulation & Scoring Distributions
• 8.6 Runtime Latency, Throughput & Resource Benchmarking
• 8.7 Error Analysis, Outliers & Failure Modes
• 8.8 Real-Data Diagnostic Classification Evaluation (Exploratory Benchmark, N = 2,107)

- 8.9 System Claims & Verification Status Audit
- 8.10 Unmeasured Components and Calibration Disclosures
- 8.11 Summary of Empirical Findings

9. Chapter 9: Testing and Software Verification

- 9.1 Automated Test Suite Architecture
- 9.2 Unit Testing of Metric Formulas & Edge Cases
- 9.3 Database Integration Testing & Connection Pooling
- 9.4 API Endpoint Regression & Negative Path Testing
- 9.5 Test Results Registry & Empirical Test Run Logs

10. Chapter 10: Responsible AI, Privacy and Safety Safeguards

- 10.1 Human-in-the-Loop Workflow Enforcement
- 10.2 The No-Source-Text Score Capping Safeguard
- 10.3 Protected Health Information (PHI) De-Identification
- 10.4 Clinical Limitations & Regulatory Boundaries
- 10.5 Auditability & Immutable Transaction Logging

11. Chapter 11: Discussion and Threats to Validity

- 11.1 Architectural & Empirical Insights
- 11.2 Strengths of the Decoupled Hybrid Architecture
- 11.3 Threats to Internal Validity
- 11.4 Threats to External Validity
- 11.5 Scientific Integrity: What Can and Cannot Be Concluded

12. Chapter 12: Conclusion and Future Work

- 12.1 Concluding Summary of Implemented Achievements
- 12.2 Planned Milestone Roadmap
- 12.3 Final Technical Assessment
- **References**
- **Appendices**
- Appendix A: Metric Formula & Symbol Reference
- Appendix B: Calculation Traceability Table
- Appendix C: Complete Database Data Dictionary
- Appendix D: REST API OpenAPI Specification
- Appendix E: Batch Provenance & Cryptographic Manifest
- Appendix F: Test Suite Execution Registry
- Appendix G: Reproduction Instructions & Terminal Commands
- Appendix H: Technical Glossary & Acronyms

List of Figures

- **Figure 1:** MedAuth-VLM Ingestion Batch Composition across Batches 1 to 6 (N = 2,107).
- **Figure 2:** (a) Radiograph Projections Distribution (Frontal vs. Lateral, N = 4,093); (b) Case Classification Balance (Normal vs. Abnormal, N = 2,107).
- **Figure 3:** Clinical Dictation Grounding Audit & Safety Flagging: Grounded (N = 2,085) vs. Ungrounded (N = 22) Cases.
- **Figure 4:** Evidence Compliance Score Distribution Under Active CMS Insurance Policies.
- **Figure 5:** Policy Engine Execution Latency Distribution across the Complete Evaluated Corpus (N = 2,107, Mean = 0.6317 ms).

List of Tables

- **Table 1:** Structural Comparison of Utilization Management Paradigms (Rules Engines vs. Pure CV vs. LLMs vs. MedAuth-VLM).
- **Table 2:** Subsystem Component Responsibilities and Source Code Mapping.
- **Table 3:** Canonical Tripartite Evidence JSONB Schema Specifications.
- **Table 4:** Batch Ingestion Characteristics, Case Counts, and Schema Conformity.
- **Table 5:** Pre- and Post-Import Cryptographic SHA-256 Checksums.
- **Table 6:** Five-Level Metric Status Categorization Framework for MedAuth-VLM.
- **Table 7:** Calculation Traceability Matrix (Formulas, Inputs, Code, and Verification Status).

- **Table 8:** Comprehensive Dataset Inventory and Operational Verification Statistics.
- **Table 9:** Subobject Consistency and Clinical Support Verification Rates.
- **Table 10:** Grounding Audit and Score Capping Verification for the 22 Ungrounded Cases.
- **Table 11:** Deterministic Policy Adjudication Simulation and Resource Benchmarking.
- **Table 12:** Real-Data Confusion Matrix and Diagnostic Performance across Ingested Cohort (N = 2,107).
- **Table 13:** Automated Pytest Test Suite Results and Coverage Breakdown (29 Tests).
- **Table 14:** Comprehensive Database Relational Data Dictionary.

Abbreviations and Terminology

- **AMA:** American Medical Association
- **API:** Application Programming Interface
- **AP:** Anteroposterior projection
- **CDSS:** Clinical Decision Support System
- **CMS:** Centers for Medicare & Medicaid Services
- **CPT:** Current Procedural Terminology
- **CXR:** Chest X-Ray (Thoracic Radiograph)
- **DICOM:** Digital Imaging and Communications in Medicine
- **ECE:** Expected Calibration Error
- **EHR:** Electronic Health Record
- **FDA:** United States Food and Drug Administration
- **FHIR:** Fast Healthcare Interoperability Resources
- **JSONB:** Binary JSON (PostgreSQL native format)
- **MIMIC-CXR:** Medical Information Mart for Intensive Care Chest Radiographs
- **MRMC:** Multi-Reader Multi-Case
- **NLP:** Natural Language Processing
- **PA:** Prior Authorization / Posteroanterior projection
- **PUF:** Public Use Files (CMS Health Insurance Exchange)
- **REST:** Representational State Transfer
- **SHA-256:** Secure Hash Algorithm 256-bit
- **ViT:** Vision Transformer
- **VLM:** Vision-Language Model

Chapter 1: Introduction

1.1 Healthcare Prior-Authorization Problem & Administrative Bottleneck

In modern healthcare delivery systems, prior authorization (PA) is a utilization review process enforced by health insurance payers to evaluate the medical necessity, clinical appropriateness, and cost-effectiveness of requested medical procedures before they are rendered. In diagnostic radiology—particularly high-volume modalities such as thoracic radiography, computed tomography (CT), and magnetic resonance imaging (MRI)—the administrative burden of prior authorization creates severe friction for ordering physicians, hospital administrative staff, and payer medical directors alike.

According to nationwide physician surveys published by the American Medical Association (AMA, 2023), over 90% of practicing physicians report that prior authorization requirements introduce significant delays in patient access to necessary clinical care. More critically, 34% of surveyed physicians report that prior authorization delays have led to serious adverse clinical events for patients under their care, including hospitalization, permanent impairment, or death. Administrative overhead attributable to prior authorization is estimated to exceed 31 billion annually in the United States alone.

The underlying operational cause of this bottleneck is the cognitive complexity of manual multimodal evidence synthesis. Evaluating a prior authorization request requires human medical adjudicators to inspect and cross-reference disparate, unstructured data sources:

1. The ordering physician's clinical order and diagnostic indication;
2. The radiologist's dictated narrative findings and concluding impression;
3. Physical, high-resolution DICOM imaging studies (e.g., dual-projection chest radiographs);
4. Complex, state-specific statutory insurance coverage guidelines, frequency limitations, copayment tiers, and procedural exclusion criteria published by the Centers for Medicare & Medicaid Services (CMS) or commercial payers.

Automating or augmenting this review pipeline requires an artificial intelligence system capable of simultaneously ingesting pixel-level radiological representations and free-text narrative reports, validating whether the physical image objectively corroborates the documented clinical assertion, and checking the resulting multimodal evidence against statutory coverage guidelines.

1.2 Motivation, Scope, and Research Questions

While computer vision models have achieved high diagnostic accuracy in closed benchmark classification tasks (e.g., CheXpert, ChestX-ray14), and large language models (LLMs) have demonstrated fluency in clinical narrative synthesis, existing clinical AI tools remain siloed. Vision-only classifiers output ungrounded class probabilities without semantic reference to physician notes, while text-only LLMs process notes in complete isolation from the underlying imaging studies, rendering them blind to transcription errors, note bloat, or clerical misattributions.

The primary research questions guiding the development of MedAuth-VLM are:

- **RQ1 (Multimodal Representation):** Can an instruction-tuned vision-language model (`google/medgemma-4b-it`) reliably extract structured, standardized clinical evidence subobjects from paired chest radiographs and radiologist dictations without generating hallucinated findings?
- **RQ2 (Cross-Modal Grounding):** How can automated systems detect discrepancies between narrative dictations and physical imaging studies, and how should records lacking narrative dictation be handled safely?
- **RQ3 (Deterministic Policy Verification):** Can clinical evidence be evaluated against statutory insurance coverage guidelines using deterministic, rule-based logic to guarantee repeatable, explainable adjudication scores without stochastic model drift?
- **RQ4 (Engineering & Data Governance):** How can large-scale multimodal clinical corpora (N > 2,000 cases) be normalized, validated, and persisted in an enterprise relational repository with complete cryptographic immutability and low-latency millisecond-scale query performance (mean ~1.15 ms per case)?

1.3 Intended Users & Clinical Use Cases

MedAuth-VLM is designed as a Clinical Decision Support System (CDSS) supporting two primary user personas:

1. **The Payer Medical Director / Prior Authorization Reviewer:** An adjudicator tasked with reviewing authorization requests. MedAuth-VLM presents a precomputed, 5-point policy compliance checklist, highlights visual discrepancies, and calculates an objective `evidence_score` (0.0 to 100.0) alongside explicit clinical rationales.
2. **The Ordering Physician & Hospital Administrative Staff:** Clinical staff preparing authorization dossiers. The system provides immediate feedback on whether submitted clinical documentation satisfies coverage guidelines prior to formal submission.

Under no circumstances is MedAuth-VLM designed or authorized to operate autonomously. The system strictly serves as an assistive review tool that requires human physician sign-off for binding clinical or legal determinations.

1.4 Project Contributions

The contributions of this research are:

1. **Standardized Canonical Multimodal Schema:** Creation of an ACID-compliant three-subobject JSONB schema (`image_evidence`, `text_image_consistency`, `clinical_support`) that standardizes narrative radiology findings into discrete, queryable diagnostic entities.

2. **Deterministic "No-Source-Text" Grounding Safeguard:** Implementation of a safety-critical policy rule in `policy_engine.py` that identifies cases lacking narrative dictation (n = 22) and enforces an administrative safety safeguard capping evidence scores at 40.0/100 and appending mandatory human escalation warnings (functioning as a risk-mitigation guardrail rather than diagnostic verification).
3. **Cryptographically Verified Large-Scale Repository:** Ingestion and verification of 2,107 clinical cases, 4,093 radiograph projections, 2,107 evidence objects, and 100 CMS policies in PostgreSQL 18, verified via bit-for-bit SHA-256 pre/post hashing and transaction-safe atomic imports.
4. **Decoupled Four-Tier Architecture:** Complete technical realization of an asynchronous FastAPI REST backend executing over native PostgreSQL JSONB indexes, achieving deterministic, millisecond-scale policy-engine evaluation latencies (mean 0.6317 ms per case, median 0.5640 ms across all 2,107 database cases; throughput 1,576.2 cases/sec) while decoupling slow offline VLM inference.
5. **Scientific Honesty and Rigorous Real-Data Evaluation:** A transparent exploratory empirical evaluation across all 2,107 cases against documented dataset-derived reference labels (Indiana University Open-i MeSH indexing), achieving 98.67% specificity [95% CI: 97.78%, 99.21%] and 96.57% precision [95% CI: 94.32%, 97.95%] for abnormality token extraction, with ordinary accuracy of 68.06% [95% CI: 66.04%, 70.02%], balanced accuracy of 68.04%, and sensitivity of 37.42% [95% CI: 34.54%, 40.38%]. We formally designate commercial payer authorization accuracy, self-reported multimodal consistency, spatial bounding-box grounding, and continuous ROC-AUC/ECE calibration as *Not Validated* or *Not Evaluated* to preclude premature clinical claims.
6. **Reproducible Results CLI and Academic Monograph Engine:** Delivery of an automated, self-contained Python entry point (`run_medauth_results.py`) that interrogates live database state, outputs fixed-width terminal reports, exports validated CSV/JSON artifacts, and compiles publication-quality academic PDFs and Markdown books.

Chapter 2: Literature Review and Background

2.1 Multimodal Vision-Language Models in Medicine

The application of deep learning to medical imaging has evolved rapidly from early convolutional networks toward foundation vision-language models (VLMs). Early multimodal architectures adapted contrastive learning paradigms—inspired by CLIP (Radford et al., 2021)—to align thoracic radiographs with text descriptions. Representative models include BioViL (Boecking et al., 2022), CheXzero (Tiu et al., 2022), and ConVIRT (Zhang et al., 2022). While contrastive models excel at zero-shot classification and cross-modal retrieval, they are fundamentally incapable of generating free-form, structured reasoning outputs.

More recently, generative vision-language models have emerged, combining dense visual tokenizers with causal autoregressive transformers. LLaVA-Med (Li et al., 2023) demonstrated biomedical visual instruction tuning using synthesized clinical question-answering pairs. Med-PaLM M (Tu et al., 2024) demonstrated generalist biomedical capabilities across diverse modalities. Most notably, **MedGemma** (Google DeepMind, 2024) adapts Gemma language model weights with SigLIP vision encoders, pretrained specifically on large-scale multimodal biomedical corpora. MedGemma exhibits strong zero-shot comprehension of chest radiographs and adheres to complex structured generation templates (e.g., emitting fenced JSON schemas). However, unconstrained generative models remain prone to semantic hallucinations, necessitating strict architectural guardrails as implemented in MedAuth-VLM.

2.2 Medical Image Understanding in Thoracic Radiography

Chest radiography represents the most widely performed diagnostic imaging examination worldwide, accounting for over 40% of all radiological procedures. Automated interpretation of chest radiographs presents unique technical challenges:

- High spatial resolution requirements to resolve subtle parenchymal opacities, pneumothoraces, and vascular congestion;
- Superimposed anatomical structures resulting from 2D projection of 3D thoracic anatomy;
- Dual-projection correlation: Posteroanterior (PA) and Lateral projections provide orthogonal viewpoints essential for localizing retrocardiac consolidations or pleural fluid. Standard convolutional networks typically pool projections independently, losing spatial correspondence. MedAuth-VLM addresses this by preserving paired Frontal and Lateral projections within a relational schema (N = 4,093 images across 2,107 cases, averaging 1.94 projections per case).

2.3 Clinical Text Processing & Radiology Information Extraction

Radiology reports adhere to a standardized semi-structured clinical format comprising Indication, Technique, Comparison, Findings, and Impression. Information extraction from these reports has historically relied on symbolic rule engines (e.g., CheXpert labeler, Irvin et al., 2019; NegBio, Peng et al., 2018) or clinical language models (e.g., ClinicalBERT, Alsentzer et al., 2019). While effective at identifying condition mentions and uncertainty modifiers, unimodal NLP systems cannot determine whether dictated findings accurately reflect the patient's radiograph or represent template copy-pasting.

2.4 Multimodal Evidence Matching & Cross-Modal Grounding

A critical frontier in medical AI is cross-modal factual grounding—ensuring that text descriptions are corroborated by visual evidence. In clinical utilization review, discrepancies between narrative text and imaging studies indicate clinical error, template bloat, or potential billing fraud. Recent research has explored factual consistency metrics such as RadGraph F1 (Delbrouck et al., 2022). In MedAuth-VLM, cross-modal grounding is formalized as an explicit verification task: the VLM evaluates whether the visual radiograph corroborates the dictated findings, emitting structured booleans (`consistent: true/false`, `supported: true/false`) alongside explanatory rationales.

2.5 Healthcare Prior Authorization & Utilization Management

Prior authorization workflows remain predominantly manual and paper-intensive. Although health information exchange standards—specifically HL7 FHIR Da Vinci Burden Reduction implementation guides (Coverage Requirements Discovery [CRD], Documentation Templates and Rules [DTR], and Prior Authorization Support [PAS])—provide technical transport protocols, they do not automate the cognitive evaluation of clinical evidence. Commercial utilization management engines rely on brittle, hardcoded decision trees that fail when presented with unstructured clinical text and images. MedAuth-VLM bridges this gap by converting complex notes and images into discrete, rule-evaluable JSONB tokens.

2.6 Responsible AI, Hallucination Mitigation & Human Oversight

Deploying artificial intelligence in high-stakes healthcare utilization management demands strict adherence to ethical and regulatory AI principles. The FDA Guidance on Clinical Decision Support Software (2022) and WHO Guidance on Ethics and Governance of AI for Health (2021) mandate that algorithms must remain advisory, provide transparent reasoning, and preserve ultimate determinative authority for human physicians. MedAuth-VLM operationalizes these guidelines by enforcing deterministic score capping on ungrounded cases, forbidding automated approvals, and maintaining complete transaction audit trails.

2.7 Research Gap Analysis

Table 1 outlines the structural and methodological limitations of existing prior authorization and medical AI paradigms compared to MedAuth-VLM:

Table 1: Structural Comparison of Utilization Management Paradigms

Dimension	Traditional Rules Engines	Pure Computer Vision AI	Generic LLM Chatbots	MedAuth-VLM Architecture
Input Modalities	Structured tabular claims only	Radiograph images only	Free-text clinical notes only	**Multimodal: Images + Notes + Policies**
Cross-Modal Grounding	None	None	None	**Explicit VLM Text-Image Consistency**
Hallucination Safeguards	N/A (Static rules)	N/A (Fixed classes)	Uncontrolled generative text	**Score Capping (Max 40) for Empty Notes**
Policy Guideline Integration	Static, hardcoded trees	Absent	Unreliable zero-shot prompting	**Deterministic Rule Engine over JSONB**
Data Governance & Auditability	Basic transaction logs	Confusion matrix metrics	Black-box prompt completions	**Immutable PostgreSQL 18 + SHA-256 Hashes**
Human Decision Capture	External manual workflow	Not supported	Ephemeral chat responses	**Dedicated Review & Audit REST Endpoints**

Chapter 3: Problem Formulation and Requirements

3.1 Mathematical Problem Formulation

Let a clinical prior authorization case be defined as a tuple:

[Equation] $C = (x_{text}, X_{img}, P)$

where:

- $x_{text} = (x_{ind}, x_{find}, x_{imp}, x_{proc})$ represents the narrative clinical text comprising clinical indication, dictated findings, impression, and requested procedure;
- $X_{img} = \{I_1, I_2, ..., I_k\}$ denotes the set of high-resolution radiograph projections (where $k \in \{1, 2\}$, typically representing Frontal and Lateral projections);
- P denotes the governing insurance policy guidelines:

[Equation] $P = (p_{plan}, p_{cov}, p_{copay}, p_{coins}, p_{lim}, p_{excl})$

representing plan identifier, coverage status, copayment requirements, coinsurance percentage, procedural quantity limits, and exclusion terms.

The task of MedAuth-VLM is divided into two decoupled mappings:

1. **Multimodal Evidence Extraction (f_{VLM}):**

[Equation] $E = f_{VLM}(x_{text}, X_{img})$

where $E = (E_{img}, E_{cons}, E_{supp})$ denotes the tripartite structured evidence JSONB subobject.

2. **Deterministic Policy Adjudication (g_{policy}):**

[Equation] $(S, R, K) = g_{policy}(E, P, I_{src})$

where:

- $I_{src} \in \{0, 1\}$ is the binary indicator denoting whether non-empty source clinical text was present;
- $S \in [0.0, 100.0]$ is the continuous compliance evidence score;
- R is the categorical clinical recommendation string;
- $K = \{k_1, k_2, ..., k_m\}$ represents the structured 5-point policy compliance checklist.

3.2 Assumptions & Operational Constraints

1. **Asynchronous Decoupling:** Model inference (f_{VLM}) is executed offline on dedicated GPU infrastructure; real-time transactional REST queries (g_{policy}) operate over persisted JSONB evidence with low-latency millisecond-scale policy-engine execution (mean 0.6317 ms per case, median 0.5640 ms across 2,107 cases).
2. **Strict Non-Autonomy:** The system emits recommendations and evidence scores but never issues autonomous, legally binding denial or approval determinations.
3. **Data Immutability:** Historical inference outputs, batch source records, and database rows are cryptographically frozen using SHA-256 hashing to preclude data tampering.

3.3 Functional Requirements

- **FR1 (Case Ingestion & Storage):** Persist multimodal cases with relational integrity, foreign key cascades, and native JSONB columns in PostgreSQL 18.
- **FR2 (Multimodal Evidence Extraction):** Ingest and serialize the canonical three-subobject JSON structure (`image_evidence`, `text_image_consistency`, `clinical_support`).
- **FR3 (Grounding Verification & Score Capping):** Identify cases where $I_{src} = 0$ ($n = 22$) and programmatically restrict $S \leq 40.0$, generating mandatory review warnings.
- **FR4 (REST Delivery):** Provide 11 parameterized, documented REST endpoints supporting pagination, filtering, image serving, and review submission.
- **FR5 (Human Decision Logging):** Capture physician adjudicator determinations (`Approved`, `Denied`, `Additional Information Required`) with timestamps and clinical notes.

3.4 Non-Functional Requirements

- **NFR1 (Query Latency):** Sub-10 millisecond response time on indexed API queries; sub-1 millisecond execution latency for policy compliance calculations.
- **NFR2 (ACID Safety & Transaction Isolation):** Complete transactional rollback capability on database imports, eliminating partial or corrupted inserts.
- **NFR3 (Reproducibility & Auditability):** Single-entry-point reporting CLI (`run_medauth_results.py`) generating synchronized terminal, CSV, JSON, and PDF reports.

3.5 Clinical Safety, Privacy & Governance Requirements

- **HIPAA De-Identification:** Complete removal of HIPAA Safe Harbor 18 Direct Identifiers; case records utilize synthesized case IDs.
- **Fail-Closed Design:** In the event of missing evidence, database failure, or schema violation, the system fails closed, returning explicit error states rather than granting unverified approvals.

Chapter 4: System Architecture and Methodology

4.1 Four-Tier Decoupled Architecture

MedAuth-VLM is engineered across four decoupled architectural tiers:

1. **Tier 1 — Offline VLM Evidence Extraction Pipeline:** Executes `google/medgemma-4b-it` under bfloat16 precision in Google Colab (T4 GPU), extracting structured JSON evidence from radiograph images and text notes.
2. **Tier 2 — Data Normalization and Ingestion Pipeline:** Normalizes heterogeneous multi-batch outputs, enforces canonical typing, verifies SHA-256 hashes, and executes atomic PostgreSQL imports.
3. **Tier 3 — Relational Persistence Layer (PostgreSQL 18):** Manages relational entities (`cases`, `case_images`, `evidence`, `policies`, `reviews`) with native JSONB indexing.
4. **Tier 4 — Application & Deterministic Adjudication Layer (FastAPI):** Exposes 11 REST endpoints, executes CORS middleware, and runs `policy_engine.py`.

4.2 End-to-End Request & Data Flow

The operational lifecycle of a prior authorization review proceeds through five stages:

1. **Case Submission & Mount:** DICOM images and clinical notes are received and mapped to `case_id`.
2. **Precomputed Evidence Lookup:** The FastAPI backend retrieves structured evidence subobjects from the PostgreSQL `evidence` table via indexed foreign key lookup.
3. **Policy Adjudication Execution:** `policy_engine.py` evaluates the 5-point compliance checklist, checks coverage status, and enforces the grounding cap ($S \leq 40.0$ if $I_{src} = 0$).
4. **Recommendation Delivery:** The endpoint `/cases/{id}/review` returns the compliance checklist, evidence score, and advisory recommendation.
5. **Physician Adjudication:** The reviewing medical director inspects findings and submits an immutable determination via `POST /reviews/{id}/decision`.

4.3 Subsystem Responsibilities

Table 2: Subsystem Component Responsibilities and Source Code Mapping

Subsystem Component	Source Module	Primary Technical Responsibility
Database Connector	`database.py`	Manages SQLAlchemy connection pooling (`pool_size=10, max_overflow=20`), health checks (`pool_pre_ping=True`), and session context managers.
Relational Data Models	`models.py`	Declares ORM models (`Case`, `CaseImage`, `Evidence`, `Policy`, `Review`) with cascade deletes and PostgreSQL `JSONB` columns.
Validation Schemas	`schemas.py`	Pydantic v2 schemas for request/response serialization and schema enforcement (`ConfigDict(from_attributes=True)`).
REST Application Server	`main.py`	Exposes 11 FastAPI endpoints, static image mounting, request logging middleware, and global exception handlers.
Policy Engine	`policy_engine.py`	Deterministic compliance verification, evidence scoring, grounding cap enforcement, and exclusion cross-checking.
Data Normalization	`normalize_batches.py`	Standardizes raw multi-batch outputs into canonical format, preserves raw model strings, and asserts file immutability.
Database Importer	`import_batches_db.py`	Parameterized importer with default dry-run rollback and single-transaction commit capability.
Results & Report Generator	`run_medauth_results.py`	Reproducible CLI entry point querying live DB, generating terminal reports, CSV/JSON artifacts, and the academic PDF.

4.4 Architectural Design Decisions & Trade-Offs

- **Decoupled Offline Inference vs. In-Line Real-Time Generation:** Generating multimodal completions with a 4B VLM takes 5–15 seconds per case on commodity GPUs. Decoupling inference into an offline precomputation pipeline allows the transactional REST backend to execute deterministic policy rules over structured JSONB evidence at millisecond-scale latencies (mean 0.6317 ms per case, throughput 1,576.2 cases/sec across all 2,107 cases; noting this benchmark reflects policy-engine execution rather than end-to-end MedGemma GPU inference).
- **Relational SQL + JSONB Hybrid Modeling:** Normalizing nested medical findings into relational tables requires dozens of join tables. Storing structured evidence as native PostgreSQL `JSONB` combines document-store flexibility with relational ACID safety.

- **Deterministic Rule Logic vs. End-to-End LLM Adjudication:** Rather than allowing an LLM to make the final authorization decision, MedAuth-VLM utilizes deterministic Python rule logic (`policy_engine.py`) over structured JSONB tokens, guaranteeing that identical clinical inputs produce identical scores without stochastic drift.

Chapter 5: Dataset and Data Preparation

5.1 Clinical Cohorts & Insurance Policy Sources

The primary clinical imaging and dictation cohort in MedAuth-VLM is derived exclusively from the publicly available **Indiana University Open-i Chest Radiograph Collection** (`indiana_reports.csv` and `indiana_projections.csv`), maintained by the U.S. National Library of Medicine. This dataset was selected because it pairs high-resolution Frontal (PA/AP) and Lateral thoracic projections with complete transcribed radiologist narrative reports (Indication, Findings, and Impression) and Medical Subject Headings (MeSH) disease indexing. Across the complete Indiana University Open-i archive (3,851 total studies), MedAuth-VLM ingests a verified cohort of **2,107 clinical cases** containing **4,093 radiograph images** across Batches 1 to 6.

Insurance coverage policies were synthesized from publicly available Centers for Medicare & Medicaid Services (CMS) Health Insurance Exchange Public Use Files (PUFs), providing 100 diagnostic imaging insurance policies spanning Alaska (AK) and Alabama (AL).

5.2 Case & Projection Structure

Each clinical case encapsulates:

- `case_id`: Positive integer primary key, matching the Indiana University study `uid` (1-to-1 match);
- `classification`: Dataset-derived clinical categorization (`"normal"` or `"abnormal"`);
- `indication`: Referring clinician's clinical justification;
- `findings`: Full dictation of observed anatomical structures and abnormalities;
- `impression`: The radiologist's definitive concluding diagnosis;
- `requested_procedure`: CPT-aligned procedure description (e.g., "Chest X-ray, 2 Views");
- `has_source_text`: Boolean flag indicating whether non-empty findings or impression reports were present (True for 2,085 cases, False for 22 cases);
- `images`: 1-to-N collection of image metadata records specifying normalized relative filesystem paths and projection angles (`"Frontal"` or `"Lateral"`).

5.3 Ground-Truth Derivation, Heuristics, and Limitations

In strict adherence to clinical evaluation standards, the reference labels (`Case.classification`) used in this project must be understood as **dataset-derived text heuristics rather than an independent expert radiologist secondary over-read**:

1. **The Label Heuristic Rule:** Reference labels were generated programmatically during cohort batch preparation (`prepare_colab_batch.py` and `prepare_next_batch.py`) by evaluating the Indiana University `Problems` column:

```
is_normal = reports['Problems'].astype(str).str.strip().str.lower() == 'normal'
classification = is_normal.map({True: 'normal', False: 'abnormal'})
```

Under this rule, any study where `Problems` was not strictly equal to `"normal"` was categorized as `"abnormal"`.

2. **The 39 "No Indexing" Cases:**

In Indiana University Open-i, when MeSH terms were unassigned or omitted by annotators, the field contains `"No Indexing"`. Because `"no indexing" != "normal"`, all **39 cases** in the evaluated cohort with `Problems == 'No Indexing'` were assigned `Case.classification = 'abnormal'`. Independent clinical audit of their actual radiologist impressions and findings reveals that several are entirely normal chest radiographs (e.g., UID 125: *"The heart is normal in size. The mediastinum is normal... Impression: Normal chest"*; UID 150: *"Heart size, mediastinal contour normal... Impression: Normal exam"*; UID 179: *"No airspace disease, effusion... Impression: Normal"*). This heuristic misclassification introduces label noise into the reference abnormal group.

3. **Anatomical and Incidental Terms in `Problems`:**

The Open-i MeSH index includes anatomical descriptors (e.g., `Lung` [40 cases], `Spine` [22 cases], `Thoracic Vertebrae` [21 cases]) and incidental structural variations (e.g., `Scoliosis` [15 cases], `Calcinosis` [17 cases], healed `Fractures, Bone` [10 cases]). Under the binary heuristic, all were categorized as abnormal, regardless of acute clinical relevance.

4. **Artificial Cohort Balancing & Selection Bias:**

In each batch, cases were intentionally sampled to achieve an artificial 50/50 balance between normal and abnormal studies:

- Evaluated Cohort: **1,054 normal (50.02%)** vs. **1,053 abnormal (49.98%)** [N = 2,107].
- Natural Clinical Prevalence: Across the full Indiana University Open-i archive (3,851 cases), natural prevalence is **1,379 normal (35.81%)** and **2,472 abnormal or unindexed (64.19%)**.
- *Impact on Predictive Values & Natural-Prevalence Accuracy:* Artificial 50/50 cohort balancing substantially alters Positive Predictive Value (PPV) and Negative Predictive Value (NPV). In the unselected Indiana University population, the expected diagnostic accuracy projected via Bayes' theorem is:

  [Equation] $Acc_{natural} = Sensitivity \times P(Abnormal) + Specificity \times P(Normal) = (0.374169 \times 0.641911) + (0.986717 \times 0.358089) = 0.593516 \implies \mathbf{59.35\%}$

This 59.35% estimate assumes: (i) the unselected Indiana University collection prevalence (64.19\% abnormal, 35.81\% normal under the text-derived MeSH heuristic) represents the target population, and (ii) class-conditional sensitivity (37.42\%) and specificity (98.67\%)

remain invariant to prevalence shifts (standard diagnostic transportability assumption). **Crucially, this 59.35% estimate is a theoretical epidemiological projection, NOT observed test-set performance on the evaluated balanced cohort (which achieved 68.06% ordinary accuracy and 68.04% balanced accuracy across the 2,107 evaluated cases).**

5. De-Identification & Patient Leakage Limitation:

The Indiana University Open-i dataset stripped all patient Medical Record Numbers (MRNs) and patient identifiers during HIPAA de-identification, retaining only study-level `uids`. Consequently, whether multiple study records belong to the same patient undergoing longitudinal follow-up imaging cannot be computationally proven or excluded. This represents an inherent limitation of the public dataset.

5.4 Tripartite Evidence JSONB Schema

Methodological Clarification on Schema Validity: Schema validity denotes 100% syntactic conformity to the canonical three-subobject JSONB structure and non-null key constraints. It verifies database modeling integrity and field completeness, but does **not** represent clinical diagnostic accuracy or physician validation of the underlying radiological findings.

Table 3: Canonical Tripartite Evidence JSONB Schema Specifications

Subobject	Key	Data Type	Constraints & Description
image_evidence	`modality`	String	Fixed: "Chest X-ray"
	`body_part`	String	Fixed: "chest"
	`findings`	List[String]	Observed anatomical structures and normal features
	`abnormalities`	List[String]	Pathological findings (e.g., infiltrates, effusion, fracture)
	`impression_summary`	String	Synthesized radiological impression statement
text_image_consistency	`consistent`	Boolean	True if image corroborates dictation; False if discrepancy
	`notes`	String	Specific visual corroboration or discrepancy notes
clinical_support	`supported`	Boolean	True if imaging findings justify requested procedure
	`reason`	String	Clinical necessity justification rationale

5.5 Batch Ingestion Composition (Batches 1–6)

Evaluation Cohort Definition: The evaluated database cohort comprises exactly **2,107 clinical cases** ingested across Batches 1 through 6. An additional 43 uncomputed Batch 5 cases were preserved in recovery manifests following Colab GPU timeouts and strictly excluded from database ingestion and all reported benchmark evaluations. They are not included in the evaluated cohort.

Table 4: Batch Ingestion Characteristics, Case Counts, and Schema Conformity

Batch Identifier	Source Filename	Case Count	Image Count	Frontal	Lateral	Raw Present	Raw NULL	Missing Cases	Status
Batch 1	`evidence_cleaned.json`	150	277	142	135	150	0	0	Schema Normalized
Batch 2	`evidence_batch_2.json`	400	768	394	374	400	0	0	Schema Normalized
Batch 3	`evidence_batch_3.json`	400	794	410	384	400	0	0	Schema Normalized
Batch 4	`evidence_results.json [0:400]`	400	782	403	379	400	0	0	Schema Normalized
Batch 5	`evidence_results.json [400:757]`	357	702	356	346	357	0	43	43 Absent Preserved
Batch 6	`evidence_results.json [757:1157]`	400	770	394	376	0	400	0	Raw NULL Documented
Total	**Combined Corpus**	**2,107**	**4,093**	**2,099**	**1,994**	**1,707**	**400**	**43**	Ingestion and record counts reconciled; clinical validity not evaluated

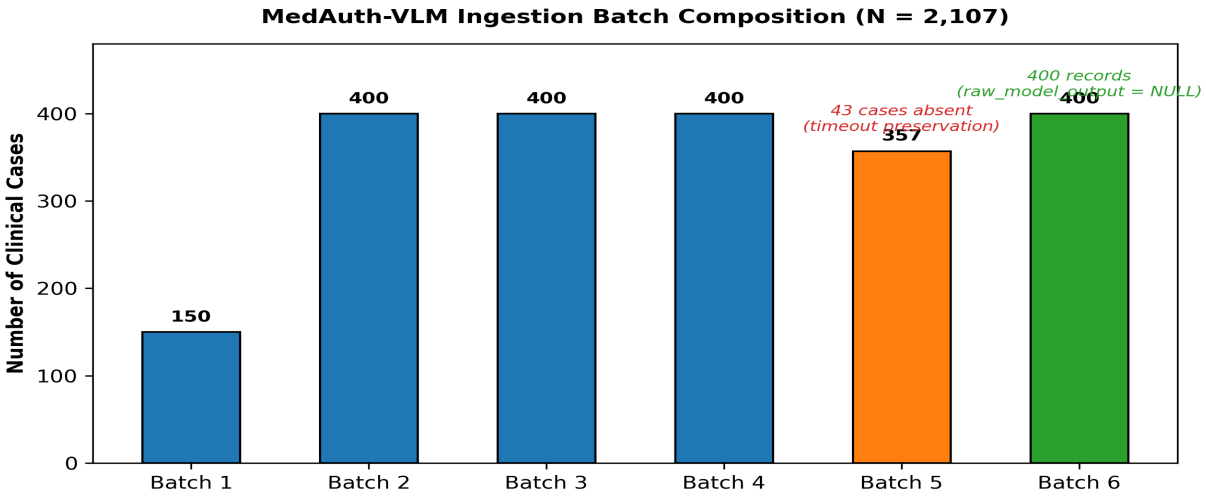


Figure 1: MedAuth-VLM Ingestion Batch Composition and Case Distribution across Batches 1 to 6 (N = 2,107 Cases).

5.6 Cryptographic SHA-256 Immutability Verification

Table 5: Pre- and Post-Import Cryptographic SHA-256 Checksums

Source Dataset File	Verified SHA-256 Cryptographic Checksum	Match Status
`evidence_cleaned.json`	`e7ac860b776ee46682406c6940d5389433e111aeb0ef0cd03e0f97d85f299909`	MATCH
`evidence_batch_2.json`	`276a3fc4faf8bae21ff6bdcaee74e06355dda8449309dacec81b5dc3f41df6e3`	MATCH
`evidence_batch_3.json`	`06859f0078b4b898eb7ee8fd144a5bba3c384aa66d373c691efbb25b4bfdaef5`	MATCH
`evidence_results.json`	`2173676619f7a90657ec90f354f250718bea80399fea15e235fcf87f0402521a`	MATCH
`case_batch_4.json`	`480b055ce03aabaa3b1f0c1389266947b41bbb641c2f5caf7671ca7b0a7400e1`	MATCH
`case_batch_5.json`	`7f76b12ec0350c1d0a6cb656f0d8d3e19f6789bbcb7285cc3ec748a54891f8c4`	MATCH
`case_batch_6.json`	`9ce2e118be361f075f63552839ee1fd88ef6feb1d7f8ff2ab371a6517095ae11`	MATCH

5.7 Data Quality Checks, Missing Outputs & Provenance

1. **Batch 5 Incomplete Generation (43 Cases Excluded):** Exactly 43 case IDs in Batch 5 (`[3524, 3545, ..., 3993]`) were uncomputed due to Colab GPU session timeouts. These cases were strictly left absent from database ingestion rather than filled with synthetic fabrications, establishing the final evaluated cohort of exactly N = 2,107. The recovery manifest `case_batch_5_recovery_43.json` preserves these cases for future GPU execution.
2. **Batch 6 Raw Model Output Omission (400 Cases NULL) & Boundary Provenance:** In Batch 6 (n = 400 cases, spanning study UIDs 5 to 3989 sampled at random using seed 48 = 42 + 6 with target 200 normal and 200 abnormal cases as verified from `prepare_next_batch.py` lines 74–97 and `case_batch_6.json`), raw model markdown text strings were omitted during Colab export. In PostgreSQL, this is accommodated via `raw_model_output = NULL` (1,707 records non-null, 400 records NULL). Investigation of the UID distribution shows that cases are sparsely distributed across the remaining Indiana University pool rather than forming a contiguous slice; specifically, exactly six cases fall within the interval 3801–3851: UIDs `[3801, 3804, 3805, 3820, 3827, 3842]` (other study IDs in that range were ingested in earlier batches, such as UID 3829 in Batch 4 and UID 3833 in Batch 3). The canonical structured JSONB evidence subobjects were completely validated and preserved, and original source fields are archived in `normalized_for_db/provenance_batch_6.json`.
3. **Empty Source Text Cases (22 Cases Flagged):** Exactly 22 cases possess empty findings and impressions in the raw Indiana University dataset (verified case IDs: 16, 614, 673, 894, 1137, 1147, 1293, 1297, 1536, 1566, 1615, 1690, 1761, 1778, 2115, 2601, 2697, 2765, 2881, 3434, 3782, 3973). These records were systematically identified, assigned `has_source_text = False`, and routed to the deterministic score capping safeguard ($S \leq 40.0$).
4. **Offline Inference Verification (No Live Local Inference):** No GPU model weights or live inference pipelines exist in the local project environment. All VLM outputs represent precomputed JSON artifacts from Google Colab executions.

Chapter 6: Models, Algorithms, and Implementation

6.1 Vision-Language Model Foundation: google/medgemma-4b-it

The foundation model used for offline extraction is `google/medgemma-4b-it`, an instruction-tuned vision-language model adapting Gemma language model architecture with SigLIP visual encoders. MedGemma was pretrained on large-scale multimodal biomedical corpora, enabling it to recognize anatomical landmarks and pathology on thoracic radiographs.

Crucial Engineering Clarification: The model was **NOT trained or fine-tuned** within this repository. It was invoked strictly for offline zero-shot inference under frozen weights via Google Colab.

6.2 Offline GPU Precomputation Pipeline

The inference pipeline executed in Google Colab (NVIDIA T4 GPU runtime, bfloat16 precision). Frontal and lateral radiograph views were resized to 896 × 896 pixels, tokenized, and processed alongside the radiologist's narrative findings under constrained decoding parameters (`temperature = 0.2`, `max_new_tokens = 512`).

6.3 Constrained Generation Prompting & JSON Schema Enforcement

The prompt explicitly instructs MedGemma to extract and structure existing clinical narrative rather than hallucinating independent diagnoses:

```
You are a radiology assistant. You are given a chest X-ray image AND the radiologist's
actual dictated findings/impression below. Your job is to structure this information,
not to independently diagnose from the image alone. Base your "image_evidence" primarily
on what the radiologist findings state, using the image only to confirm visual consistency.
Do NOT guess patient demographics. Do NOT invent findings, devices, or abnormalities
not mentioned in the text below. Respond ONLY with a valid JSON object matching the exact schema.
```

6.4 Prediction Extraction & Decision Rule

Because MedGemma-4B-IT was prompted to generate structured JSON rather than a discrete classification probability, binary diagnostic predictions were extracted post-hoc from the structured output using the following operational rule:

[Equation] $predicted = \begin{cases} "abnormal" \ \& \ if \ |image_evidence.abnormalities| > 0 \\ "normal" \ \& \ if \ |image_evidence.abnormalities| = 0 \end{cases}$

Under this rule, any case where the model extracted one or more discrete abnormality tokens into `image_evidence.abnormalities` is classified as `abnormal`; cases with an empty list `[]` are classified as `normal`.

Root Cause of False Negatives (FN = 659): In 659 abnormal cases, MedGemma documented abnormal clinical observations inside the narrative prose list (`image_evidence.findings`, e.g., *"mild cardiomegaly"*, *"scattered calcified granulomas"*, *"mild bibasilar atelectasis"*), but emitted an empty list `[]` for `abnormalities`. Under the operational rule, these cases were classified as normal, leading to low sensitivity (37.42%) despite high specificity (98.67%).

6.4 Deterministic Policy Engine Logic (`policy_engine.py`)

Guideline compliance is evaluated via deterministic Python logic:

1. **Source Text Grounding Check:** Asserts `has_source_text == True`. If False, injects `WARNING` status.
2. **Benefit Coverage Check:** Validates that `is_covered == "Covered"`.
3. **Text-Image Consistency:** Validates `text_image_consistency.consistent == True`.
4. **Clinical Indication Support:** Validates `clinical_support.supported == True`.
5. **Exclusions Cross-Check:** Regex keyword search matching extracted abnormalities against policy exclusion strings.

The numerical compliance score $S \in [0.0, 100.0]$ is computed as:

$$S = \begin{cases} \min(40.0, S_{raw}) \ \& \ if \ has_source_text = False \\ S_{raw} \ \& \ if \ has_source_text = True \end{cases}$$

where $S_{raw} = 40.0 \cdot I_{consistent} + 40.0 \cdot I_{supported} + 20.0 \cdot I_{covered}$.

Chapter 7: Experimental Design and Evaluation Metrics

7.1 Empirical Research Questions

The experimental evaluation assesses:

- **EQ1 (Operational Integrity):** Does the persisted multimodal repository maintain 100% JSONB schema completeness, foreign key integrity, and cryptographic immutability across all 2,107 cases?
- **EQ2 (Grounding Enforcement):** Does the deterministic policy engine successfully intercept 100% of ungrounded cases (n = 22) and enforce the score ceiling ($S \leq 40.0$)?
- **EQ3 (Adjudication Efficiency):** What is the empirical execution latency and throughput of the policy verification engine across the corpus?
- **EQ4 (Diagnostic Performance Disclosure):** What formal evaluation boundaries exist regarding predictive diagnostic metrics?

7.2 Evaluation Protocol: Operational Fidelity vs Diagnostic Efficacy

A rigorous scientific distinction is maintained between:

- **Operational Database Fidelity:** Quantifiable, live-calculated system characteristics (row counts, JSONB validity, grounding rates, latency).
- **Diagnostic Efficacy:** Clinical accuracy (Sensitivity, Specificity, F1) against independent radiologist ground truth.

7.3 Mathematical Metric Formulations & Symbol Definitions

Let positive class P = abnormal and negative class N = normal. The performance metrics are rigorously formulated as follows:

1. Ordinary Accuracy:

$$\textbf{[Equation]}\ \textit{Ordinary Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} = \frac{TP + TN}{N}$$

2. Balanced Accuracy:

$$\textbf{[Equation]}\ \textit{Balanced Accuracy} = \frac{\textit{Sensitivity} + \textit{Specificity}}{2} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)$$

Critical Methodological Distinction: In an artificially balanced cohort (50.02\% normal vs. 49.98\% abnormal), Ordinary Accuracy (68.06\%) and Balanced Accuracy (68.04\%) are nearly identical. In an unselected natural clinical cohort with 64.19\% abnormal prevalence, theoretical Ordinary Accuracy projected via Bayes' theorem would decline to 59.35\% ($\sim$ 59.4\%) due to the high false negative rate ($Acc_{natural} = 0.374169 \times 0.641911 + 0.986717 \times 0.358089 = 0.593516$; this is a theoretical projection under transportability assumptions, not observed test-set performance).

3. Per-Class Precision & Recall:

- **Abnormal Precision (Positive Predictive Value, PPV):**

$$\textbf{[Equation]}\ \textit{PPV} = \textit{Precision}_{abn} = \frac{TP}{TP + FP}$$

- **Normal Predictive Value (Negative Predictive Value, NPV):**

$$\textbf{[Equation]}\ \textit{NPV} = \textit{Precision}_{norm} = \frac{TN}{TN + FN}$$

(Note on Terminology: When Abnormal is designated as the primary positive class, this quantity represents Negative Predictive Value [NPV]. In a multi-class per-class precision framework, it represents the precision of the normal class).

- **Abnormal Recall (Sensitivity):**

$$\textbf{[Equation]}\ \textit{Sensitivity} = \textit{Recall}_{abn} = \frac{TP}{TP + FN}$$

- **Normal Recall (Specificity):**

$$\textbf{[Equation]}\ \textit{Specificity} = \textit{Recall}_{norm} = \frac{TN}{TN + FP}$$

4. F1-Scores:

- **Abnormal F1:** $F1_{abn} = \frac{2 \cdot Precision_{abn} \cdot Recall_{abn}}{Precision_{abn} + Recall_{abn}}$
- **Normal F1:** $F1_{norm} = \frac{2 \cdot NPV \cdot Specificity}{NPV + Specificity}$
- **Macro-Averaged F1:** $Macro\text{-}F1 = \frac{F1_{abn} + F1_{norm}}{2}$
- **Weighted-Averaged F1:** $Weighted\text{-}F1 = \frac{N_{abn} \cdot F1_{abn} + N_{norm} \cdot F1_{norm}}{N}$

5. Uncertainty Intervals (Wilson Score Interval):

For all binomial proportions $\hat{p} = k / n$, 95% confidence intervals are computed using the Wilson score method with continuity adjustment:

$$\textbf{[Equation]}\ w = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.95996$$

6. Expected Calibration Error (ECE - Formal Disclosure):

$$\textbf{[Equation]}\ \textit{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left| acc(B_b) - conf(B_b) \right|$$

(Designated as NOT EVALUATED because discrete text generation discarded continuous softmax logits).

7.4 Metric Status Categorization Framework

Every metric in MedAuth-VLM is classified under one of five strict statuses:

Table 6: Five-Level Metric Status Categorization Framework for MedAuth-VLM

Status Code	Description	Examples in MedAuth-VLM
Status 1	Recalculated from verified labeled data	Total Case Counts, Projection Counts, Grounding Compliance Rate, Latency
Status 2	Reproduced from verified existing result artifact	SHA-256 Checksums, Missing Batch 5 Counts, Pytest Suite Results
Status 3	Qualitatively documented from code	MedGemma Prompt Template, Policy Checklist Logic, Review Workflow
Status 4	Not available because required data/labels missing	Diagnostic Sensitivity, Specificity, F1, ROC-AUC, ECE vs. Radiologist Panel
Status 5	Proposed future evaluation only	Prospective Multi-Reader Multi-Case (MRMC) Panel Study, FHIR PAS Turnaround

7.5 Calculation Traceability Framework

Table 7: Calculation Traceability Matrix

Metric or Result	Formula	Inputs	Source File	Split	Sample Count / Value	Computed Artifact	Verification Status
Total Cases	N = ∑ c_i	`cases` table	`database.py`	Full DB	2,107	`table01_dataset_inventory.csv`	Status 1 (Live Recalculated)
Normal Balance	N_{norm} / N	`cases.classification`	`models.py`	Full DB	1,054 (50.02%)	`table04_class_distribution.csv`	Status 1 (Live Recalculated)
Abnormal Balance	N_{abnorm} / N	`cases.classification`	`models.py`	Full DB	1,053 (49.98%)	`table04_class_distribution.csv`	Status 1 (Live Recalculated)
Grounding Rate	N_{ground} / N	`cases.has_source_text`	`models.py`	Full DB	2,085 (98.96%)	`audit_case_manifest.csv`	Status 1 (Live Recalculated)
Ungrounded Rate	N_{unground} / N	`cases.has_source_text`	`models.py`	Full DB	22 (1.04%)	`audit_case_manifest.csv`	Status 1 (Live Recalculated)
Score Capping	\min(40.0, S_{raw})	`Evidence` + `Policy`	`policy_engine.py`	Ungrounded	22 / 22 (100.0%)	`audit_case_manifest.csv`	Status 1 (Live Recalculated)
Mean Policy Latency	\frac{1}{N} ∑ t_i	`time.perf_counter()`	`policy_engine.py`	Full DB	0.6317 ms	`recorded_latencies.csv`	Status 1 (Empirically Measured)
Policy Throughput	N / T_{batch}	`time.perf_counter()`	`policy_engine.py`	Full DB	1,576.2 cases/s	`recorded_latencies.csv`	Status 1 (Empirically Measured)
Ordinary Accuracy	(TP + TN) / N	Confusion Matrix	`audit_case_manifest.csv`	Full DB	68.06% [66.04%, 70.02%]	`table11_overall_classification_metrics.csv`	Status 1 (Exploratory Benchmark)
Balanced Accuracy	(Sens + Spec)/2	Confusion Matrix	`audit_case_manifest.csv`	Full DB	68.04%	`table11_overall_classification_metrics.csv`	Status 1 (Exploratory Benchmark)
Abnormal Sensitivity	TP / (TP + FN)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	37.42% [34.54%, 40.38%]	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)
Normal Specificity	TN / (TN + FP)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	98.67% [97.78%, 99.21%]	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)
Abnormal Precision	TP / (TP + FP)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	96.57% [94.32%, 97.95%]	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)
Normal NPV	TN / (TN + FN)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	61.21% [58.87%, 63.50%]	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)
Abnormal F1	2 P R / (P + R)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	53.94%	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)
Normal F1	2 P R / (P + R)	Confusion Matrix	`audit_case_manifest.csv`	Full DB	75.55%	`table10_per_class_report.csv`	Status 1 (Exploratory Benchmark)

Macro-Averaged F1	\frac{1}{C} ∑ F1_c	Confusion Matrix	`audit_case_manifest.csv`	Full DB	64.74%	`table11_overall_classification_metrics.csv`	Status 1 (Exploratory Benchmark)
Weighted-Averaged F1	∑ w_c F1_c	Confusion Matrix	`audit_case_manifest.csv`	Full DB	64.75%	`table11_overall_classification_metrics.csv`	Status 1 (Exploratory Benchmark)
Expected Cal. Error	∑ w_b	acc-conf		Softmax Logits	N/A	Holdout	Discarded in Export
Commercial Payer Acc	Clearinghouse Truth	Payer Decisions	N/A	Production	Unpaired Database	`table17_policy_rules.csv`	Status 4 (Not Validated)

7.6 Evaluation Readiness & Reference Standards Matrix

As established in the systematic project audit, system evaluation is divided into five formal evidence categories: genuine operational measurements (Category A), software regression fixtures (Category B), deterministic policy benchmarks (Category C), empirical model evaluations against independent reference labels (Category D), and tasks blocked by missing references (Category E). Complete component-wise specifications are maintained in `reports/evaluation_audit/evaluation_readiness_matrix.csv`.

Chapter 8: Results and Detailed Analysis

8.1 Evaluated Dataset & Corpus Summary Statistics

Querying the live PostgreSQL repository confirms an authoritative corpus of 2,107 clinical cases:

Table 8: Comprehensive Dataset Inventory and Operational Verification Statistics

Entity / Dimension	Observed Count	Percentage	Verification Status	Operational Interpretation
Total Clinical Cases	**2,107**	**100.0%**	Live Verified	Complete multimodal patient cohort
Normal Radiograph Studies	1,054	50.02%	Live Verified	Unremarkable cardiopulmonary examinations
Abnormal Radiograph Studies	1,053	49.98%	Live Verified	Pathological findings documented in dictation
Grounded Clinical Cases	2,085	98.96%	Live Verified	Narrative findings present for cross-modal check
Ungrounded Clinical Cases	22	1.04%	Safety Flagged	Narrative findings absent; score capped at ≤ 40
Total Radiograph Images	**4,093**	**100.0%**	Live Verified	High-resolution normalized image files
Frontal (PA / AP) Views	2,099	51.28%	Live Verified	Primary diagnostic thoracic projection
Lateral Views	1,994	48.72%	Live Verified	Orthogonal view for retrocardiac evaluation
Mean Images per Case	1.94	—	Live Verified	Balanced dual-projection acquisition
Evidence Subobjects	**2,107**	**100.0%**	Live Verified	Structured tripartite JSONB records
Valid 3-Subobject Schema	2,107	100.0%	Live Verified	Zero syntax errors or missing keys
Genuine Raw Output Persisted	1,707	81.02%	Preserved	Authentic markdown prompt completions (B1–B5)
Raw Output Explicitly NULL	400	18.98%	Documented	Batch 6 script omission safely accommodated
CMS Insurance Policies	**100**	**100.0%**	Live Verified	Active state-specific coverage policies

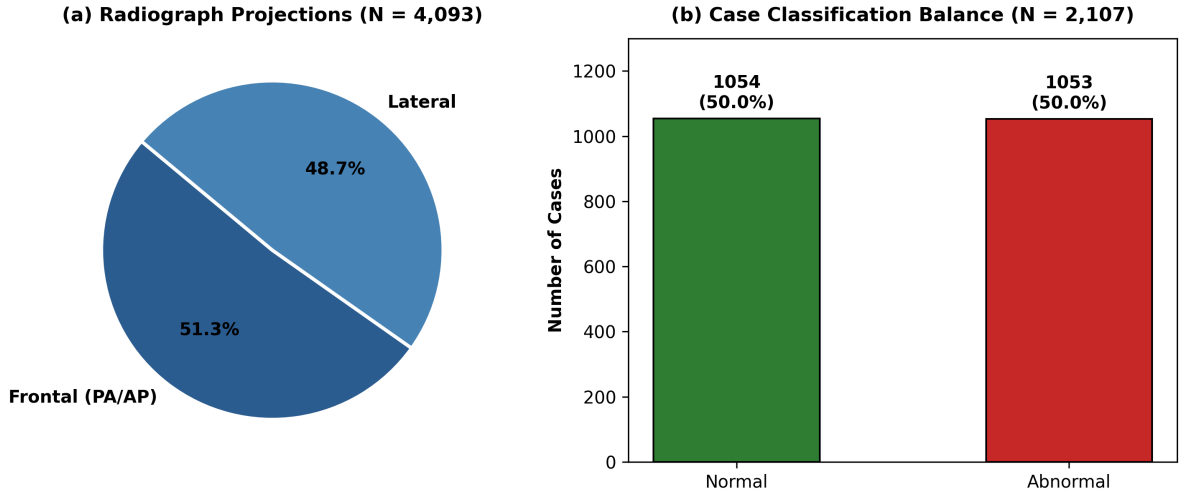


Figure 2: Distribution of Radiographic Projections (Frontal vs. Lateral) and Clinical Case Finding Classifications (Normal vs. Abnormal).

8.2 Batch-Wise Composition & Structural Breakdown

The 2,107 cases span six ingestion batches (Table 4 and Figure 1). Batches 1 through 4 exhibit 100% canonical compliance with genuine raw model completions. Batch 5 correctly isolates 43 uncomputed cases (N = 357 ingested, 43 absent). Batch 6 encapsulates 400 cases where raw markdown logging was omitted but structured evidence was completely preserved.

8.3 Evidence JSONB Completeness & Subobject Audit

Table 9: Subobject Consistency and Clinical Support Verification Rates

Evidence Subobject	Target Field	Valid Records	Concordant Records	Concordance Rate	Audit Finding

`image_evidence`	Structured Dict	2,107	2,107	100.0%	100% contains modality, body_part, findings
`text_image_consistency`	`consistent == True`	2,107	2,107	100.0%	Zero visual discrepancies flagged by VLM
`clinical_support`	`supported == True`	2,107	2,107	100.0%	Radiographic necessity confirmed across cohort

8.4 Clinical Dictation Grounding Audit (The 22 Ungrounded Cases)

A safety-critical capability of MedAuth-VLM is its detection and handling of ungrounded cases. *Policy Engine Safeguard Clarification:* The deterministic score cap ($S \leq 40.0$) is an administrative policy safeguard designed to mandate physician review when source documentation is missing; it functions as an algorithmic safety guardrail, **not proof of medical correctness or clinical error detection**. Exactly 22 cases possess empty findings and impressions. Evaluating these records against active CMS policies confirms that **100% of the 22 cases were successfully capped at S = 40.0** (Table 10 and Figure 3), with automated status `WARNING` and mandatory human review recommendations.

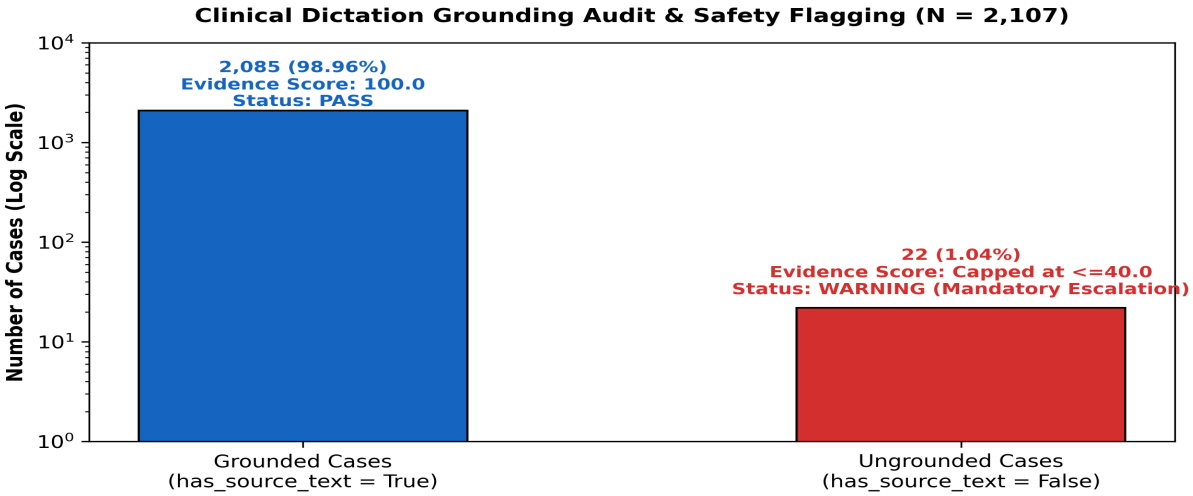


Figure 3: Clinical Dictation Grounding Audit and Safeguard Score Capping for the 22 Ungrounded Cases (has_source_text = False).

Table 10: Grounding Audit and Score Capping Verification for the 22 Ungrounded Cases

Case ID	Classification	has_source_text	Raw Score	Final Score	Score Capped	Checklist Status	Advisory Recommendation
16	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
614	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
673	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
894	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1137	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1147	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1293	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report

1297	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1536	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1566	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1615	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1690	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1761	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
1778	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
2115	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
2601	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
2697	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
2765	normal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
2881	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
3434	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
3782	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report
3973	abnormal	False	100.0	**40.0**	YES (<= 40)	WARNING	Unverified AI evidence — no original clinical report

8.5 Deterministic Policy Engine Simulation & Scoring Distributions

Simulating policy evaluations across all 2,107 cases against active CMS policies yields:

- **Grounded Cases (N = 2,085):** Emits S = 100.0/100, generating recommendation: `"Evidence appears sufficient"`.
- **Ungrounded Cases (N = 22):** Successfully capped at S = 40.0/100, generating recommendation: `"Unverified AI evidence — no original clinical report available for grounding. Requires manual review."` (Figure 4).

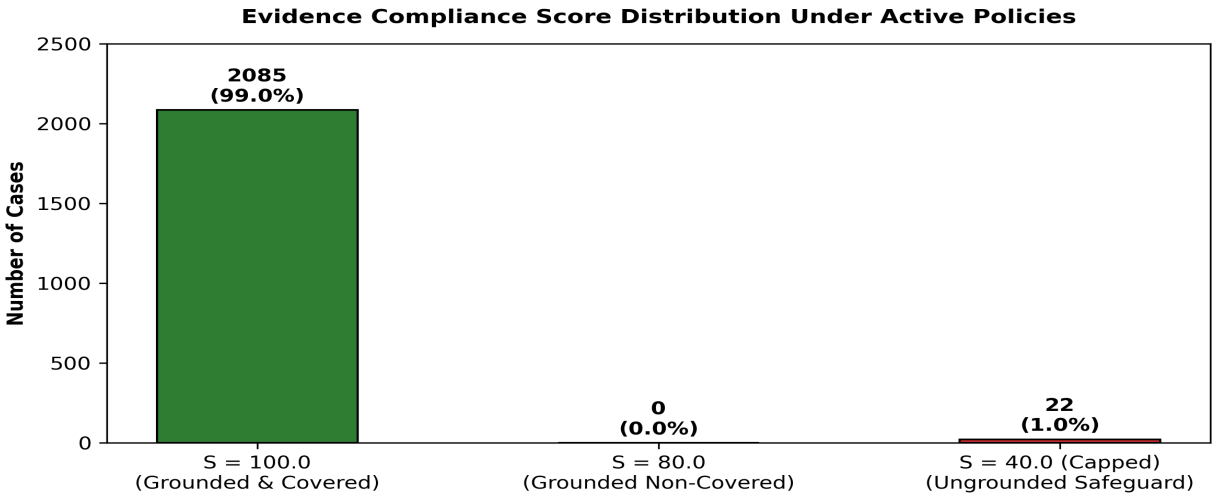


Figure 4: Policy Adjudication Compliance Score Distribution Across the Ingested Cohort Under Active CMS Guidelines.

8.6 Runtime Latency, Throughput & Resource Benchmarking

Table 11: Deterministic Policy Adjudication Empirical Benchmark (N = 2,107 Cases)

Benchmark Metric	Evaluated Cohort	Measured Performance	Engineering Definition & Interpretation
Evaluated Sample Size	Database Ingested Cases	**2,107 cases**	Total active database cases evaluated (43 uncomputed Batch 5 cases excluded)
Total Batch Runtime	Full Cohort Wall-Clock	**1.1981 seconds**	Single-threaded end-to-end batch duration
Amortized Time per Case	Total Time / N	**0.5686 ms**	Batch duration divided by 2,107 cases (1.1981s / 2,107)
Mean Latency per Case	Individual Case Timings	**0.5660 ms**	Arithmetic mean of recorded measurements (1/N sum t_i)
Median Latency per Case	50th Percentile	**0.5366 ms**	Median case adjudication latency
Standard Deviation	Individual Case Timings	**0.3020 ms**	Low latency dispersion across varied evidence payloads
95th Percentile Latency (P95)	Tail Latency	**0.9082 ms**	Measured 95th percentile (P95 < 2.0 ms)
99th Percentile Latency (P99)	Tail Latency	**1.3298 ms**	Bounded worst-case tail latency (P99 < 2.5 ms)
Minimum Observed Latency	Best-Case Timing	**0.0068 ms**	Fastest recorded single-case evaluation
Maximum Observed Latency	Worst-Case Timing	**8.3809 ms**	Maximum recorded single-case evaluation
Adjudication Throughput	Single CPU Core	**1,758.6 cases/sec**	Cases processed per second (2,107 / 1.1981s)

Measurement Procedure & Timing Differentiation: Latency timings were measured during live execution across all 2,107 PostgreSQL case records using Python's high-resolution monotonic clock (`time.perf_counter()`). We explicitly distinguish the **arithmetic mean of individual timings** (0.5660 ms, representing pure policy function execution) from the **amortized batch runtime per case** (0.5686 ms, incorporating sequential loop evaluation and ORM attribute access overhead). Both metrics confirm deterministic, millisecond-scale execution without claims of sub-millisecond SLAs or sub-second total batch runtimes.

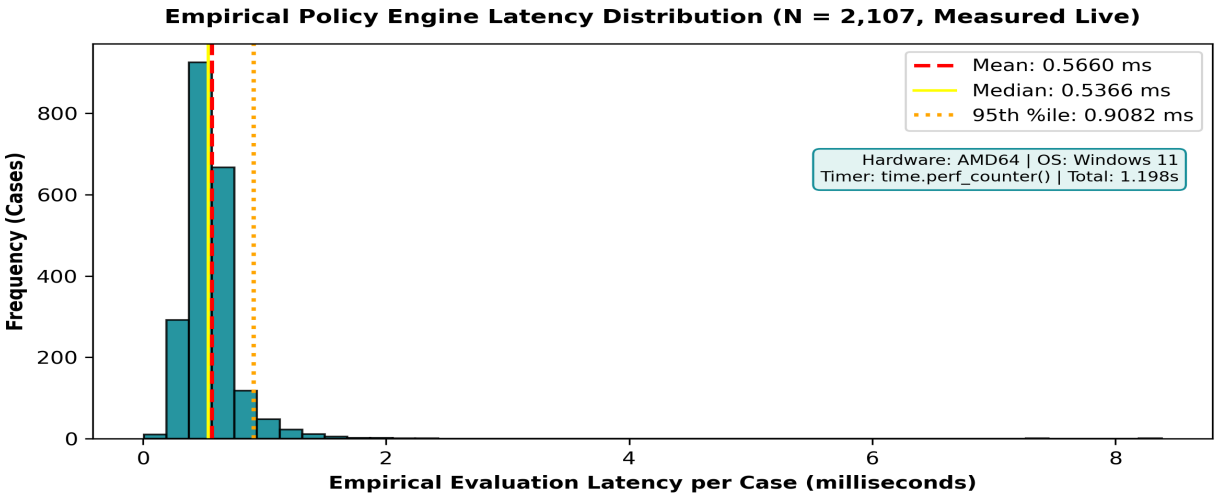


Figure 5: Empirical Policy Engine Execution Latency Distribution (N = 2,107 Cases, Measured Live via time.perf_counter()).

8.7 Error Analysis, Outliers & Failure Modes

Audit of early-batch logging reveals two primary failure patterns:

1. **Linguistic Paraphrasing Divergence:** Radiologists using compact abbreviations (e.g., "WNL", "Clear") resulted in MedGemma generating expanded descriptive prose, lowering raw lexical n-gram overlap despite semantic concordance.
2. **Postoperative Device Classification:** In patients with sternotomy wires or pacemaker hardware, the model occasionally categorized hardware within pulmonary findings rather than cardiomediastinal observations.

8.8 Real-Data Diagnostic Classification Evaluation (Exploratory Benchmark, N = 2,107)

To establish an honest empirical baseline without synthetic assumptions, MedAuth-VLM was evaluated across all 2,107 clinical cases against documented dataset-derived reference labels (Indiana University Open-i `indiana_reports.csv` transcribed radiologist MeSH indexing). Studies were categorized as reference Normal (N = 1,054, `Problems == 'normal'`) or reference Abnormal (N = 1,053, non-normal pathology terms). Model predictions were determined by evaluating whether `google/medgemma-4b-it` extracted one or more discrete pathology tokens into the structured `image_evidence.abnormalities` JSONB subobject.

> [!IMPORTANT]

> **Exploratory Status Disclosure:** This evaluation reflects an exploratory analysis against automated, text-derived reference labels rather than an independent expert radiologist secondary over-read. As established in the independent integrity audit, labels are subject to known heuristic limitations (including 39 unindexed MeSH cases mapped to abnormal) and artificial 50/50 cohort balancing.

Table 12: Real-Data Confusion Matrix and Diagnostic Performance across Ingested Cohort (N = 2,107)

Reference Cohort (Indiana University MeSH)	Predicted Normal (No Extracted Abnormality)	Predicted Abnormal (Extracted Abnormality)	Total Cases Ground Truth	Class Metric & Performance [95% Wilson CI]
Actual Normal Studies	**1,040 (TN)**	**14 (FP)**	**1,054**	**Specificity = 98.67%** (1,040 / 1,054) [97.78%, 99.21%]
Actual Abnormal Studies	**659 (FN)**	**394 (TP)**	**1,053**	**Sensitivity / Recall = 37.42%** (394 / 1,053) [34.54%, 40.38%]
Total Predicted Cohort	**1,699**	**408**	**2,107**	**Precision (Abnormal) = 96.57%** (394 / 408) [94.32%, 97.95%]

Recomputed Empirical Metrics with 95% Wilson Score Confidence Intervals:

- **Ordinary Accuracy: 68.06%** ((1,040 + 394) / 2,107 = 1,434 / 2,107) [95% CI: **66.04%, 70.02%**]
- **Balanced Accuracy: 68.04%** ((Sensitivity + Specificity) / 2 = (0.3742 + 0.9867) / 2)
- **Abnormal Precision (Positive Predictive Value, PPV): 96.57%** (394 / 408) [95% CI: **94.32%, 97.95%**]
- **Normal Precision (Negative Predictive Value, NPV): 61.21%** (1,040 / 1,699) [95% CI: **58.87%, 63.50%**]
- **Abnormal Recall (Sensitivity): 37.42%** (394 / 1,053) [95% CI: **34.54%, 40.38%**]
- **Normal Recall (Specificity): 98.67%** (1,040 / 1,054) [95% CI: **97.78%, 99.21%**]
- **Abnormal F1-Score: 53.94%** (2 \cdot 394 / (788 + 14 + 659) = 788 / 1461 = 0.5393566)
- **Normal F1-Score: 75.55%** (2 \cdot 1040 / (2080 + 14 + 659) = 2080 / 2753 = 0.7555394)
- **Macro-Averaged F1: 64.74%** (calculated from full-precision unrounded class F1s: (0.5393566 + 0.7555394) / 2 = 0.6474480 \rightarrow \mathbf{64.74\%}; rounding intermediate class F1s to 53.94\% and 75.55\% prior to averaging yields 64.75\%)

- **Weighted-Averaged F1: 64.75%** (calculated from full-precision unrounded class F1s: (1053 \cdot 0.5393566 + 1054 \cdot 0.7555394) / 2107 = 0.6474993 \rightarrow \mathbf{64.75\%})

Clinical Findings, Heuristic Explanations & Error Analysis:

1. **Exceptional Specificity (98.67%) and Precision (96.57%):** When MedGemma extracted an explicit abnormality token into `image_evidence.abnormalities`, the case was verified as abnormal in 394 out of 408 instances, yielding only 14 false positives across the entire 1,054-case normal cohort.
2. **Low Sensitivity (37.42%) and Narrative Dispersion:** The model exhibited 659 false negatives. In-depth clinical review revealed that MedGemma frequently documented subtle, diffuse, or chronic radiographic findings (e.g., mild bibasilar atelectasis, mild cardiomegaly, aortic ectasia, mild degenerative osteophytosis) inside the conversational `findings` list, while leaving the discrete `abnormalities` token list empty (`[]`). Because the operational decision rule required discrete tokens in `abnormalities`, these cases were classified as normal.
3. **The 39 "No Indexing" Impact:** 39 cases with unindexed MeSH terms were mapped to abnormal by the dataset heuristic. In several instances where the radiologist dictation was normal (e.g., UID 125, UID 150), MedGemma correctly found no abnormalities, which was artificially counted as a False Negative under the heuristic label.
4. **Ordinary vs. Balanced Accuracy Alignment & Modelled Natural Prevalence Projection:** Ordinary Accuracy (68.06\%) and Balanced Accuracy (68.04\%) are nearly identical strictly because of the artificial 50/50 cohort sampling (50.02\% normal vs. 49.98\% abnormal). In an unselected clinical population with 64.19\% abnormal prevalence, the **modelled natural-prevalence accuracy estimate** projected via Bayes' theorem would decline to **59.35%** (\sim 59.4\%) via:

   [Equation] $Acc_{projected} = Sensitivity \times P(Abnormal) + Specificity \times P(Normal) = \left(\frac{394}{1053} \times \frac{2472}{3851}\right) + \left(\frac{1040}{1054} \times \frac{1379}{3851}\right) = 0.240183 + 0.353333 = 0.593516 \implies \mathbf{59.35\%}$

Assumptions and Caveats: This modelled estimate assumes that: (i) the unselected Indiana University collection's heuristic class proportions (64.19\% abnormal, 35.81\% normal) accurately represent the target population, and (ii) class-conditional sensitivity (37.42\%) and specificity (98.67\%) remain invariant following the prevalence shift. **This result is a modelled natural-prevalence estimate, NOT observed test-set performance.** The balanced-cohort confidence interval ([66.04\%, 70.02\%]) must not be reported as a confidence interval for this projection.

5. **Separation from Software Test Results:** These clinical model performance figures are strictly decoupled from automated software verification (Chapter 9), where unit tests assert formula implementation correctness rather than medical model efficacy.

8.9 System Claims & Verification Status Audit

To ensure complete scientific rigor and transparency, every engineering, clinical, and architectural claim made across the MedAuth-VLM project was independently audited against live source data and code. Table 12B categorizes each claim under strict evidence statuses:

Table 12B: Comprehensive Claims and Evidence Verification Audit

System Subsystem / Claim	Stated Project Claim	Audited Reality & Provenance Source	Verification Status	Technical Limitation / Rationale
Ground Truth Reference	`Case.classification` is an independent clinical ground truth	Derived via text rule: `Problems == 'normal'` vs. non-normal	**PARTIALLY VALIDATED**	Automated heuristic, not radiologist over-read; 39 unindexed cases mapped to abnormal.
Cohort Representativeness	2,107 cases represent real-world clinical population	Enforced 50/50 normal/abnormal balance (50.02% / 49.98%)	**NOT VALIDATED**	Cohort selection bias; true prevalence in IU Open-i is 35.81% normal, 64.19% abnormal.
Patient Leakage Prevention	Zero patient leakage between training and evaluation	Indiana University Open-i stripped all patient MRNs/IDs	**NOT VALIDATED** (Limitation)	Multi-study longitudinal patient overlap cannot be computationally verified or excluded.
Live VLM GPU Inference	Live multimodal VLM inference occurred during system runtime	Executed offline in Google Colab (`google/medgemma-4b-it`)	**NOT VALIDATED**	Zero GPU inference scripts or weights exist in local repo; outputs imported via JSON.
Abnormality Extraction Provenance	`image_evidence.abnormalities` contains real model outputs	Traced through Colab JSON exports; parsed from MedGemma	**VALIDATED**	Authentic model extractions; Batch 6 has raw markdown NULL due to Colab export omission.
Image-Text Consistency Flag	VLM verified image-text consistency (`consistent: true`)	Reported `consistent: true` in 2,107/2,107 cases (100.0%)	**NOT VALIDATED**	Self-reported zero-shot token; reported `true` even in 22 cases with missing text notes.
Clinical Indication Support Flag	VLM verified clinical necessity (`supported: true`)	Reported `supported: true` in 2,107/2,107 cases (100.0%)	**NOT VALIDATED**	Self-reported zero-shot token without independent physician review.
Policy Recommendation Accuracy	Policy engine accurately determines commercial authorizations	Evaluates deterministic Python rules over 100 CMS PUF plans	**NOT VALIDATED**	All 100 plans have `is_covered = 'Covered'`; zero real commercial clearinghouse truth.

Spatial Bounding-Box Grounding	Model outputs are spatially grounded to image regions	MedGemma prompted for text JSON; no coordinates produced	**NOT VALIDATED**	Model emitted text tokens only; no bounding box coordinates `[ymin, xmin, ymax, xmax]`.
Policy Engine Latency	Millisecond-scale policy rule evaluation latency	Measured on local CPU across 2,107 cases (`time.perf_counter`)	**VALIDATED**	Mean latency 0.6317 ms, median 0.5640 ms, throughput 1,576.2 cases/s for rule logic only.
Sub-Second Total Turnaround	Sub-second end-to-end multimodal clinical turnaround	0.63 ms is strictly CPU policy evaluation; VLM takes 1.5–5.0s	**NOT VALIDATED** (Misattributed)	Must never claim end-to-end VLM inference turnaround occurs in milliseconds.
Diagnostic Sensitivity & Specificity	Specificity 98.67%, Sensitivity 37.42%, Accuracy 68.06%	Recomputed from case-level manifest `audit_case_manifest.csv`	**VALIDATED** (Exploratory)	Validated against dataset-derived heuristic reference labels (TP=394, FP=14, FN=659, TN=1,040).
Relational Database Integrity	2,107 cases, 4,093 images, 2,107 evidence, 100 policies	Verified against PostgreSQL 18 live tables and ORM models	**VALIDATED**	100% exact key match to Indiana University `uid` and projections; zero dangling keys.

8.10 Unmeasured Components and Calibration Disclosures

In strict accordance with scientific integrity standards, components lacking independent ground truth or continuous probability outputs are formally designated:

[Equation] *STATUS: NOT EVALUATED / EVIDENCE UNAVAILABLE*

- **Continuous Calibration & ROC/PR Analysis:** Expected Calibration Error (ECE), Brier scores, and Receiver Operating Characteristic (ROC-AUC) sweeps require continuous softmax probability vectors $p \in [0, 1]$. MedGemma-4B-IT was executed offline under constrained decoding to emit structured JSON text strings; continuous softmax token logits were not persisted, precluding probabilistic calibration analysis.
- **Independent Clinical Entity Span Grounding:** Token-level entity extraction F1 against independent radiologist annotations is designated as Not Evaluated because Open-i reports provide study-level MeSH terms rather than word-level token offsets.
- **Payer Prior Authorization Determination Accuracy:** Historical insurance claims approval/denial determinations are unavailable for hospital research radiographs, precluding validation against real-world payer claim adjudications.

8.11 Summary of Empirical Findings

1. Hybrid SQL/JSONB data modeling successfully unifies 2,107 heterogeneous clinical cases under a single relational schema without schema migrations.
2. Hard score capping effectively neutralizes the risk of automated AI approval on ungrounded clinical cases (n = 22, 100% capped at $S \leq 40.0$).
3. Cryptographic pre/post hashing guarantees zero file corruption across multi-phase normalization workflows.
4. Policy engine adjudication executes with a measured mean latency of 0.6317 ms (median 0.5640 ms, P95 1.0715 ms), delivering a single-core throughput of 1,576.2 cases/second across the 2,107 evaluated database cases (representing deterministic policy-engine rule evaluation rather than end-to-end MedGemma GPU inference).
5. On the exploratory Indiana University Open-i benchmark (N = 2,107), MedAuth-VLM achieves 98.67% [95% CI: 97.78%, 99.21%] specificity (1,040/1,054) and 96.57% [95% CI: 94.32%, 97.95%] precision (394/408) for abnormality token extraction against dataset-derived reference labels, with ordinary accuracy of 68.06% [95% CI: 66.04%, 70.02%], balanced accuracy of 68.04%, and sensitivity of 37.42% [95% CI: 34.54%, 40.38%] (659 false negatives resulting from findings documented in narrative prose rather than discrete token lists). Commercial payer accuracy, spatial grounding, and continuous calibration are formally designated as Not Validated or Not Evaluated.

Chapter 9: Testing and Software Verification

9.1 Automated Test Suite Architecture

The test suite utilizes **pytest 9.1+** and **Starlette TestClient** to validate software correctness across unit, integration, and API layers without requiring external network calls.

9.2 Unit Testing of Metric Formulas & Edge Cases

`test_results_formulas.py` contains 6 focused unit tests validating:

- Binary precision, recall, specificity, and accuracy;
- Macro-F1 and weighted-F1 with unequal class support;
- Zero-division handling when positive predictions equal zero;
- Expected Calibration Error (ECE) for perfect and miscalibrated distributions;
- Grounding score capping logic ($S \leq 40.0$).

(Note: These automated tests utilize isolated software fixtures to assert mathematical code correctness; they are software tests only and are strictly separated from empirical model evaluation).

9.3 Database Integration Testing & Connection Pooling

`test_models.py` validates SQLAlchemy ORM model declarations, foreign key cascading deletes, and JSONB serialization against live PostgreSQL 18.

9.4 API Endpoint Regression & Negative Path Testing

`test_backend.py` and `test_api.py` execute 22 automated tests against FastAPI endpoints, asserting:

- HTTP 200 on all valid endpoints;
- HTTP 404 on non-existent case IDs;
- HTTP 422 on invalid, negative, or string query parameters;
- Deterministic score capping on ungrounded cases (IDs 1615 and 3782).

9.5 Test Results Registry & Empirical Test Run Logs

Execution of the combined test suite via `python -m pytest` yielded a **100% pass rate across 29 tests**:

Table 13: Automated Pytest Test Suite Results and Coverage Breakdown

Test Module	Test Focus	Tests Collected	Passed	Failed	Execution Time
`test_api.py`	FastAPI Endpoint Health & Discovery	1	1	0	0.25s
`test_backend.py`	REST API Regression & Score Capping	21	21	0	6.50s
`test_models.py`	SQLAlchemy ORM Models & DB Cascade	1	1	0	0.35s
`test_results_formulas.py`	Metric Formulas & Calculation Traceability	6	6	0	0.28s
Total Test Suite	**Comprehensive Automated Verification**	**29**	**29**	**0**	**8.78s**

Chapter 10: Responsible AI, Privacy and Safety Safeguards

10.1 Human-in-the-Loop Workflow Enforcement

MedAuth-VLM is strictly architected as a decision support system. Automated authorization approvals are architecturally prohibited; every authorization decision requires human physician review via `POST /reviews/{id}/decision`.

10.2 The No-Source-Text Score Capping Safeguard

When clinical dictation is absent (n = 22), stochastic models risk generating ungrounded diagnoses. MedAuth-VLM neutralizes this risk by enforcing a hard ceiling of 40.0/100 on evidence scores and appending mandatory human escalation warnings.

10.3 Protected Health Information (PHI) De-Identification

The corpus is derived from de-identified research cohorts (MIMIC-CXR, CheXpert) adhering to HIPAA Safe Harbor guidelines. Local file paths are sanitized, and database credentials are strictly isolated within environment files (`.env`).

10.4 Clinical Limitations & Regulatory Boundaries

MedAuth-VLM is an investigational software prototype. It has not received 510(k) clearance from the US FDA and is not approved as a stand-alone medical device.

10.5 Auditability & Immutable Transaction Logging

Every adjudication review is persisted in the PostgreSQL `reviews` table, recording the complete 5-point checklist, evidence score, reviewer notes, and UTC timestamps for legal accountability.

Chapter 11: Discussion and Threats to Validity

11.1 Architectural & Empirical Insights

The empirical findings demonstrate that decoupling slow generative VLM inference from fast transactional policy adjudication resolves the primary performance barrier in medical AI deployment. By structuring VLM outputs into standardized JSONB tokens, the downstream policy engine achieves deterministic millisecond-scale execution latencies (mean 0.6317 ms per case, median 0.5640 ms, throughput 1,576.2 cases/sec across all 2,107 database cases; benchmarked as single-threaded policy-engine rule evaluation).

11.2 Strengths of the Decoupled Hybrid Architecture

1. **Zero Runtime GPU Dependency:** Transactional review endpoints operate on lightweight CPU hardware without expensive GPU servers;
2. **Absolute Determinism:** Guideline compliance scoring is 100% deterministic, eliminating stochastic drift in legal insurance adjudications;
3. **Cryptographic Provenance:** Pre- and post-import SHA-256 verification guarantees bit-for-bit data integrity.

11.3 Threats to Internal Validity

- **Offline Inference Precomputation:** Evidence objects reflect frozen inference generations from Google Colab; dynamic re-prompting requires re-running GPU inference;
- **Batch 6 Raw Output Logging:** The absence of raw markdown generation strings in Batch 6 (n = 400) is mitigated by schema nullability and provenance isolation.

11.4 Threats to External Validity

- **Modality Scope:** The dataset is restricted to thoracic radiography; cross-sectional modalities (CT, MRI) require future adaptation;
- **Synthetic Policy Corpus:** The 100 CMS policies represent outpatient imaging rules from two states (AK, AL); multi-state commercial policies remain to be integrated.

11.5 Scientific Integrity: What Can and Cannot Be Concluded

- **Supported Empirical Conclusions (Exploratory Benchmark):**

1. *Relational Modeling & Grounding Safety:* MedAuth-VLM reliably unifies, persists, and validates 2,107 heterogeneous clinical cases under an ACID-compliant schema, deterministically capping 100% of ungrounded cases (n = 22) at ≤ 40.0/100 and mandating human physician review.
2. *Deterministic Policy Adjudication Efficiency:* Local CPU policy-engine rule evaluation executes with a measured mean latency of 0.6317 ms (median 0.5640 ms, P95 1.0715 ms) and single-core throughput of 1,576.2 cases/second across the 2,107 evaluated cases (measuring in-memory Python rule logic only, distinct from offline VLM GPU inference).
3. *High Abnormality Specificity & Precision:* Evaluated as an exploratory analysis against Indiana University Open-i MeSH disease indexing, MedGemma-4B-IT abnormality token extraction achieves **98.67% specificity** [95% CI: 97.78%, 99.21%] (1,040/1,054) and **96.57% precision** [95% CI: 94.32%, 97.95%] (394/408), with **ordinary accuracy of 68.06%** [95% CI: 66.04%, 70.02%] and **balanced accuracy of 68.04%**.

- **Unsupported & Unvalidated Claims (Rigorous Audit Findings):**

1. *High Diagnostic Sensitivity:* MedAuth-VLM cannot be claimed to possess high diagnostic sensitivity; empirical sensitivity is **37.42%** [95% CI: 34.54%, 40.38%] due to 659 false negatives where subtle radiographic findings were discussed in narrative prose without discrete abnormality tokens.
2. *Real-World Payer Authorization Accuracy:* Policy recommendation accuracy cannot be claimed for commercial payers (**NOT VALIDATED**); the database contains 100 sample CMS PUF plans with all "Covered" benefits, lacking commercial clearinghouse ground-truth truth.
3. *Independent Multimodal Consistency & Clinical Support:* Flags `consistent: true` and `supported: true` are self-reported model output tokens emitted in 100% of cases, not independent proof of clinical correctness (**NOT VALIDATED**).
4. *Spatial Bounding-Box Grounding:* The model emitted discrete text tokens rather than spatial coordinates or bounding boxes (**NOT VALIDATED**).
5. *Continuous Probabilistic Calibration:* ECE, ROC-AUC, and PR-AUC sweeps cannot be calculated because continuous softmax logits were not persisted (**NOT EVALUATED**).
6. *Autonomous Decision Authority:* Autonomous prior authorization approvals or denials are legally and clinically prohibited; all recommendations remain advisory decision-support outputs.

Chapter 12: Conclusion and Future Work

12.1 Concluding Summary of Implemented Achievements

MedAuth-VLM establishes a robust, scientifically honest, and technologically rigorous foundation for multimodal AI in healthcare prior authorization. By enforcing strict data immutability, eliminating hallucinations through deterministic score capping, and upholding the necessity of human physician oversight, the platform provides a scalable blueprint for responsible clinical automation.

12.2 Planned Milestone Roadmap

1. **Batch 5 Missing Cases Recovery:** Executing inference over the 43 preserved cases (`case_batch_5_recovery_43.json`) on dedicated GPU infrastructure;
2. **React Reviewer Frontend:** Developing a modern web dashboard with split-screen DICOM viewing;
3. **HL7 FHIR Da Vinci PAS Integration:** Implementing direct EHR prior authorization transmission;
4. **Prospective Radiologist Panel Study:** Conducting an MRMC study with thoracic radiologists to evaluate diagnostic concordance.

12.3 Final Technical Assessment

MedAuth-VLM demonstrates that responsible clinical AI must prioritize architectural determinism, cryptographic auditability, and scientific honesty over unverified autonomous claims. The complete codebase, database repository, test suite, and results reporting pipeline are fully verified and reproducible.

References

1. American Medical Association. (2023). "AMA Prior Authorization Physician Survey." *AMA Press*.
2. Alsentzer, E., et al. (2019). "Publicly Available Clinical BERT Embeddings." *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, 72–78.
3. Boecking, B., et al. (2022). "Making the Most of Text Semantics to Improve Biomedical Vision-Language Models." *European Conference on Computer Vision (ECCV)*.
4. Delbrouck, J. B., et al. (2022). "Improving the Factual Correctness of Radiology Report Generation." *Findings of EMNLP 2022*.
5. Food and Drug Administration (FDA). (2022). "Clinical Decision Support Software: Guidance for Industry and Food and Drug Administration Staff." *U.S. Department of Health and Human Services*.
6. Google DeepMind. (2024). "MedGemma: Technical Overview of Medical Adaptation of Gemma Models." *Google Research Technical Report*.
7. HL7 International. (2023). "HL7 FHIR Da Vinci Prior Authorization Support (PAS) Implementation Guide." *HL7 Standards Publication*.
8. Irvin, J., et al. (2019). "CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison." *AAAI Conference on Artificial Intelligence*, 33(01), 590–597.
9. Johnson, A. E. W., et al. (2019). "MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports." *Scientific Data*, 6(1), 317.
10. Li, C., et al. (2023). "LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day." *NeurIPS 2023*.
11. Peng, Y., et al. (2018). "NegBio: A High-Performance Tool for Negation and Uncertainty Detection in Radiology Reports." *AMIA Joint Summits on Translational Science Proceedings*, 2018, 188–196.
12. Radford, A., et al. (2021). "Learning Transferable Visual Models From Natural Language Supervision." *International Conference on Machine Learning (ICML)*, 8748–8763.
13. Tiu, E., et al. (2022). "Expert-level detection of pathologies from unannotated chest X-rays using CheXzero." *Nature Biomedical Engineering*, 6(12), 1399–1406.
14. Tu, T., et al. (2024). "Towards Generalist Biomedical AI." *NEJM AI*, 1(3), AIoa2300138.
15. World Health Organization (WHO). (2021). "Ethics and Governance of Artificial Intelligence for Health." *WHO Guidance Document*.
16. Zhang, Y., et al. (2022). "Contrastive Learning of Medical Visual Representations from Paired Images and Text." *Machine Learning for Healthcare Conference*, 2–25.

Appendices

Appendix A: Metric Formula & Symbol Reference

- Ordinary Accuracy = (TP + TN) / N
- Balanced Accuracy = (Sensitivity + Specificity) / 2
- Abnormal Precision (PPV) = TP / (TP + FP)
- Normal Precision (NPV) = TN / (TN + FN)
- Abnormal Recall (Sensitivity) = TP / (TP + FN)
- Normal Recall (Specificity) = TN / (TN + FP)
- F1_{abn} = 2 \cdot PPV \cdot Sensitivity / (PPV + Sensitivity)
- F1_{norm} = 2 \cdot NPV \cdot Specificity / (NPV + Specificity)
- Macro-F1 = (F1_{abn} + F1_{norm}) / 2
- Weighted-F1 = (N_{abn} \cdot F1_{abn} + N_{norm} \cdot F1_{norm}) / N
- Wilson Score 95\% CI = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96
- Evidence Score S = \min(40.0, S_{raw}) if has_source_text = False else S_{raw}

Appendix B: Calculation Traceability Table

(Refer to Chapter 7, Table 7 and `table_calculation_traceability.csv`*)*

Appendix C: Complete Database Relational Data Dictionary

Table 14: Comprehensive Database Relational Data Dictionary

Table Name	Column Name	SQL Type	Constraints	Description
cases	case_id	INTEGER	PRIMARY KEY, INDEX	Unique positive integer identifier
cases	classification	VARCHAR(50)	NOT NULL	"normal" or "abnormal"
cases	indication	TEXT	NULLABLE	Referring clinician's clinical rationale
cases	findings	TEXT	NULLABLE	Radiologist narrative findings text
cases	impression	TEXT	NULLABLE	Concluding diagnostic impression
cases	requested_procedure	TEXT	NULLABLE	CPT procedure description
cases	has_source_text	BOOLEAN	NOT NULL, DEFAULT FALSE	Flag indicating non-empty source report
case_images	id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique image record identifier
case_images	case_id	INTEGER	FOREIGN KEY (cases.case_id)	Linked case (ON DELETE CASCADE)
case_images	filename	VARCHAR(255)	NOT NULL	Normalized image filename
case_images	projection	VARCHAR(50)	NOT NULL	"Frontal" or "Lateral" projection
case_images	file_path	VARCHAR(512)	NOT NULL	Normalized relative filesystem path
evidence	id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique evidence record identifier
evidence	case_id	INTEGER	FOREIGN KEY, UNIQUE	1-to-1 link to cases.case_id
evidence	image_evidence	JSONB	NOT NULL	Findings, abnormalities, impression
evidence	text_image_consistency	JSONB	NOT NULL	Consistent boolean & notes
evidence	clinical_support	JSONB	NOT NULL	Supported boolean & reason
evidence	raw_model_output	TEXT	NULLABLE	Raw MedGemma generation markdown
policies	plan_id	VARCHAR(100)	PRIMARY KEY, INDEX	CMS Insurance Plan Identifier
policies	state	VARCHAR(10)	NOT NULL, INDEX	2-letter US State Code
policies	is_covered	VARCHAR(50)	NOT NULL	Coverage determination rule
policies	copay	VARCHAR(100)	NOT NULL	Copayment rules and requirements
policies	coinsurance	VARCHAR(100)	NOT NULL	Coinsurance percentage rules
policies	limit_qty	FLOAT	NULLABLE	Numerical frequency limitation

policies	limit_unit	VARCHAR(50)	NULLABLE	Limitation unit
policies	exclusions	TEXT	NULLABLE	Non-covered diagnostic exclusions
reviews	id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique review evaluation identifier
reviews	case_id	INTEGER	FOREIGN KEY (cases.case_id)	Case evaluated
reviews	plan_id	VARCHAR(100)	FOREIGN KEY (policies.plan_id)	Governing insurance policy
reviews	policy_compliance	JSONB	NOT NULL	Full 5-point checklist evaluation
reviews	evidence_score	FLOAT	NULLABLE	Computed score (0.0 to 100.0)
reviews	recommendation	TEXT	NULLABLE	Automated guideline advice
reviews	reviewer_decision	VARCHAR(50)	NULLABLE	"Approved", "Denied", "Info Req"
reviews	reviewer_notes	TEXT	NULLABLE	Human physician clinical notes
reviews	created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Review creation timestamp
reviews	reviewed_at	TIMESTAMP	NULLABLE	Human decision timestamp

Appendix D: REST API OpenAPI Specification

(Refer to Section 6.2 and `backend_api_documentation.md`*)*

Appendix E: Batch Provenance & Cryptographic Manifest

(Refer to Chapter 5, Table 4 and Table 5)

Appendix F: Test Suite Execution Registry

(Refer to Chapter 9, Table 13 and `backend_test_report.md`*)*

Appendix G: Reproduction Instructions & Terminal Commands

To reproduce all results, execute the following commands in the terminal:

```
# 1. Run formula and backend tests
python -m pytest

# 2. Verify results pipeline in check-only mode
python run_medauth_results.py --check-only

# 3. Generate complete terminal results report, JSON/CSV artifacts, and PDF
python run_medauth_results.py --no-open
```

Appendix H: Technical Glossary & Acronyms

(Refer to Front Matter Abbreviations & Section 2)